



Short communication

Characterization of a novel theta-type plasmid pSM409 of *Enterococcus faecium* RME isolated from raw milk

Tawsif Ahmed Kazi^a, Suranjita Mitra^a, Bidhan Chandra Mukhopadhyay^a, Sukhendu Mandal^b, Swadesh Ranjan Biswas^{a,*}

^a Department of Botany, Visva-Bharati University, Santiniketan 731235, West Bengal, India

^b Laboratory of Molecular Bacteriology, Department of Microbiology, University of Calcutta, 35, Ballygunge Circular Road, Kolkata 700019, India

ARTICLE INFO

Keywords:

Plasmid

Enterococcus faecium

Theta-replication

Iterons

ABSTRACT

Enterococcal plasmids have generated renewed interest for their indispensable role in pathogenesis and dissemination of multidrug-resistance. Recently, a novel plasmid pSM409 (4303-bp, GC% = 33.6%), devoid of antibiotic-resistance and virulence genes, has been identified in *Enterococcus faecium* RME, isolated from raw milk by us. pSM409 contains six open reading frames encoding a replication initiator protein (RepB) and five accessory proteins: antitoxin epsilon, bacteriocin immunity protein, HsdS, and two hypothetical proteins. Comparative sequence analysis of pSM409 reveals a mosaic pattern of similarity with different loci obtained from different theta plasmids, which dictates the plasmid to be heterogeneous or mosaic, possibly due to recombination. The pSM409 comprised of a typical theta-type origin of replication with four and a half direct repeats (iterons) of 22 nucleotides. The pSM409-RepB shared 76–82% homology with the RepB of reported theta plasmids from different genera, with dissimilarities mostly in its DNA-binding and C-terminal domain. The RepB sequence-based phylogenetic tree revealed its distinct position relative to the reported ones. The RepB grouped in the same clade has identical DNA-binding domains and their cognate iterons, possibly due to their sequence-specific interaction to initiate plasmid replication. Comparative analysis of the pSM409-iteron reveals that the repeats markedly differed from their closest homologues. This clade-specific relationship provides a new concept of classifying theta plasmids. The theta-type replicon identified in pSM409 has been found to be unique to *E. faecium* RME, prompting us to further investigate its utility as a vector for genetic manipulation of enterococci for health and industry.

1. Introduction

Enterococcus faecium, a member of lactic acid bacteria, is a commensal inhabiting in the gastrointestinal tract of humans and animals (Carasi et al., 2014; Murray, 1990). The bacterium is also found in a variety of food products such as milk, meat, sausage, vegetables and cheese (Yuksel et al., 2015) and certain strains have long been used in the food industry as probiotics (Franz et al., 2011; Holzapfel et al., 2018; Wu et al., 2019) or as starter cultures to produce dairy and non-dairy fermented foods (Abouelnaga et al., 2016; Gaglio et al., 2016). It is reported that most strains of this species are resistant to virtually all antibiotics including the commonly used vancomycin, gentamicin and kanamycin (Ben Braïek and Smaoui, 2019; Klibi et al., 2015). They cause

nosocomial infections such as endocarditis, bacteremia, infections in the urinary tract and the central nervous system (Abouelnaga et al., 2016; Camargo et al., 2014; Özmen Toğay et al., 2010). The widespread of antibiotic resistance in enterococci (García-Solache and Rice, 2019; Gilmore et al., 2013) impose a challenge to the treatment of the enterococcal-infections. The virulence and resistance genes have been found on both their genomes and plasmids (Arias et al., 2010; Ike, 2017; Tomita et al., 2003, 2002). Considering the enormous impact of pathogenic *E. faecium* on human health, enterococcal plasmids have been a focus of recent research in understanding pathogenicity, virulence, and evolution of antibiotic resistance. The native plasmids that do not contain transferrable virulence or resistant genes are a promising target for designing tools for genetic improvement in starter and probiotic

Abbreviations: ORF, Open Reading Frame; BLAST, Basic Local Alignment Search Tool; NCBI, National Centre for Biotechnology Information; Ori, Origin of Replication.

* Corresponding author.

E-mail address: swadesh_s2000@yahoo.co.in (S. Ranjan Biswas).

<https://doi.org/10.1016/j.gene.2021.145459>

Received 29 October 2020; Received in revised form 2 January 2021; Accepted 20 January 2021

Available online 27 January 2021

0378-1119/© 2021 Elsevier B.V. All rights reserved.

strains of enterococci and also for virulence-attenuation of pathogenic enterococci. Plasmids are of frequent occurrence in *E. faecium*. To date, 818 enterococcal plasmids have been sequenced of which 689 are from *E. faecium* (<https://www.ncbi.nlm.nih.gov/genome/browse#!/plasmids/>). Very few of these plasmids have been systematically characterized. While many native plasmids from lactic acid bacteria, such as lactobacilli and lactococci, have been exploited for the development of food-grade cloning and expression vectors (Émond et al., 2001; Lee et al., 2007; Mierau and Kleerebezem, 2005); such tools for *E. faecium* are still limited (Lee et al., 2019; Özmen Toğay et al., 2010). Therefore, a continuous search for endogenous cryptic enterococcal plasmids and their characterization is essential as the first step for vector construction.

A replicon is the foremost important region in the plasmid-based vectors. In general, there are two types of plasmid replicon, theta and rolling-circle (RCR) replicon, based on their mode of replication. The minimal replicon of a theta-replicating plasmid is composed of an AT-rich *ori* region associated with a gene coding for a replication initiator protein. Theta-plasmid replication initiates with the melting of the parental strands, followed by the continuous and discontinuous synthesis of leading and lagging strand, respectively (del Solar et al., 1998). While, the customary constituent of RCR plasmids are replication initiator (Rep) protein, the double strand origin (*dso*), and the single strand origin (*ss*). Unidirectional and uncoupled synthesis of the leading and lagging strands are the two main stages of rolling circle replication (Espinosa et al., 1995).

Interestingly, in contrast to the theta-replicating plasmids, RCR plasmids from Gram-positive bacteria often have a broad-host range and high copy number, due to the lack of host dependency, thus making them the first choice for the vector construction, but with a ssDNA replication intermediate, they have low structural and segregational stability, causing it inappropriate for the construction of expression or drug delivery system (del Solar et al., 1998; Machón et al., 2015). RCR plasmids are highly disseminable, thus posing a risk of spreading the plasmid-encoded traits or resistance to the undesired organisms. Thus, the theta-replicons have received greater attention in the field of genetic engineering with broad applications in food, pharma and nutraceutical industries. In this study, we have sequenced and characterized a novel plasmid pSM409 from *E. faecium* RME isolated from raw milk.

2. Materials and methods

2.1. Isolation and identification of lactic acid bacteria harbouring plasmid:

For the isolation of lactic acid bacteria harbouring plasmid, fresh raw milk samples were randomly collected from Raiganj, India (25°37'6.71" N and 88°7'32.1" E), brought to the laboratory in ice and serially diluted in water up to 10^{-8} before spreading onto Mann Rogosa Sharpe (MRS, Hi Media, India) agar plate and incubation for overnight at 37 °C in order to get single colonies. To obtain pure culture, morphologically distinct colonies from each plate were then repeatedly streaked on MRS plate. The single colonies were then grown in MRS broth. An overnight grown cultures of the isolates were inoculated at 2% in 5 ml broth for growth and plasmid isolation using Qiagen Spin Miniprep Kit (Qiagen, Hilden, Germany), as per manufacturer's instructions. The plasmid preparations were checked on agarose gel. For identification of the plasmid-bearing isolates to species-level, genomic DNA was isolated from the overnight-grown culture of *E. faecium* by lysozyme–proteinase K procedure and 16S rRNA gene was PCR amplified using bacteria-specific universal primers: 27F 5'-AGAGTTTGATCATGGCTC-3' and 1327R 5'-CTAGCGATTCCGACTTCA-3' following the procedure described previously (Mitra et al., 2005). The amplicons were purified through gel extraction kit (Qiagen) and sequenced by Xcelris Labs Limited, India. The nucleotide sequence obtained was used to determine homology using BLASTn program at NCBI site (Altschul et al., 1990). The 16S rRNA gene sequence of *E. faecium* RME was deposited in GenBank under

the accession number MT078711.

2.2. Plasmid sequencing:

The paired-end sequencing library of the plasmid was prepared using QIAseq FX DNA Library kit. The library preparation process was initiated with 200 ng plasmid DNA. The DNA was enzymatically cleaved into smaller fragments followed by a continuous step of end-repair where an 'A' is added to the 3' ends making the DNA fragments ready for platform-specific adapters ligation at both ends of the DNA fragments. A high-fidelity amplification step was performed using HiFi PCR Master Mix to ensure maximum yields from limited amounts of starting material. The amplified library was analysed using High Sensitivity (HS) DNA chip as per manufacturer's instructions in Bioanalyzer 2100 (Agilent Technologies) and finally the libraries were sequenced on Illumina platform (2 × 150 bp chemistry).

2.3. Assembly, analysis and confirmation of plasmid sequence:

The raw sequence data were processed to obtain high quality clean reads using Trimmomatic v0.35 using the default parameters as described previously (Bolger et al., 2014). The high-quality reads obtained were *de novo* assembled using SPADES 3.12.0 assembler (K-mer = 89), following the pre-set plasmid algorithm of SPADES (Bankevich et al., 2012) and Velvet v1.2.10 (Zerbino and Birney, 2008). The assembled sequence statistics was calculated using an in-house Perl script. The gap-filling of the assembled plasmid sequence was done by SSPACE- STANDARD -3.0 using default parameters (Boetzer et al., 2011). The coding sequences were predicted using Prodigal program at DFAST web server (<https://dfast.nig.ac.jp/>). The assembled sequence was annotated using DFAST web server and Rapid Annotations using Subsystems Technology (RAST) server (Aziz et al., 2008). Due to the presence of chromosomal sequence in the assembly, the genomic reads were filtered out using Bowtie 2 version 2.3.4.1 (Langmead and Salzberg, 2012) considering *Enterococcus faecium* Com15 chromosome (CP025022.1) as reference (based on 16S rRNA gene similarity) and the unassembled reads were reassembled *de novo* and annotated (vide supra). The open reading frames (ORFs) predicted, were manually curated and analysed further using MegaBLAST against the NCBI database, and in the case of replication protein, against the ACLAME database (<http://aclame.ulb.ac.be/>). The GC content along the whole plasmid sequence was analysed using GC Content Calculator (<https://www.biologicscorp.com/tools/GCContent/>). Complete map of the sequenced plasmid was drawn using WebDSV (<http://www.molbiotools.com/WebDSV/index.html>). The circularity of the plasmid was confirmed by restriction digestion using *EcoRI* and *NcoI*. Plasmid specific primers pSM409-F: TTTAATCACTTTTATAGCAGCGTTGT and pSM409-R: CAAATGGGTGTACCGGTTAATT were designed to confirm the plasmid assembly. PCR was carried out for 35 cycles and the PCR conditions were as follows: initial denaturation at 95 °C for 5 min followed by 35 cycles of denaturation step at 95 °C for 30 s, a primer annealing step at 52 °C for 30 s, and an extension step at 72 °C for 30 s, followed by a final extension step at 72 °C for 5 min. Conserved domain search tool (<https://www.ncbi.nlm.nih.gov/Structure/cdd/>) of NCBI and InterPro (<https://www.ebi.ac.uk/interpro/>) were used to predict the domain architecture of the replication protein. The novelty of the plasmid backbone was assessed upon visualizing sequence similarity through Circos plot generation using Circoletto (<http://tools.bat.infspire.org/circoletto/>) with the retrieved sequence, sharing homology with pSM409 upon BLASTn search using MegaBLAST algorithm of NCBI.

tBLASTn search of the replication protein of pSM409 was carried out and sequences receiving top scores were retrieved from the GeneBank database. Multiple sequence alignments were done using ClustalW of the BioEdit software package (<http://www.mbio.ncsu.edu/BioEdit>) and web-based Clustal Omega program (<https://www.ebi.ac.uk/Tools/msa/clustalo/>). The evolutionary history of the replication protein was

inferred by Neighbour-Joining method (Saitou and Nei, 1987) using MEGAX software package (Kumar et al., 2018). To assess the robustness of the trees, we performed 1000 bootstrap replication (Felsenstein, 1985). Branches corresponding to partitions reproduced in less than 50% bootstrap replicates are collapsed. The evolutionary distances were computed using the Poisson correction method (Zuckerlandl and Pauling, 1965). A total of 46 amino acid sequences were retrieved from the top tBLASTn hits of pSM409-RepB protein, for the construction of phylogenetic tree.

Plasmid origin of replication, *ori*, was annotated manually based on the knowledge of sequence conservation for the origin of replication of reported theta plasmids and was further confirmed by multiple sequence alignment of the 22-bp direct repeats (iterons) of pSM409 with the closely related *ori* sequences retrieved from the plasmid sequences, obtained from tBLASTn search of pSM409-RepB.

3. Results

3.1. Isolation and identification of the plasmid bearing isolate

Plasmids were prepared from the pure cultures of each of the isolates and screened for the presence of plasmid through agarose gel electrophoresis. One of the five isolates was found to harbour a single plasmid. The bacterial isolate containing the plasmid was identified by PCR amplification of 16S rRNA gene followed by the sequencing of the PCR amplicon. The sequence analysis revealed that the isolates are closely related with the strain *Enterococcus faecium* Com15 (CP025022.1), *Enterococcus faecium* Gr17 (CP033376.1) and *Enterococcus faecium* SRCM103470 (CP035220.1) with 16S rDNA sequence similarity of 100%, 99.9% and 99.9%, respectively. Hence, it was designated as *E. faecium* RME.

3.2. Nucleotide sequence analysis of plasmid pSM409 from *E. faecium* RME

The complete annotated-sequence of the plasmid pSM409 were deposited in GenBank under the accession number MT096402.

Sequence analysis of the plasmid pSM409 of *E. faecium* RME indicated that the plasmid consists of 4303 bp with an average GC-content of 33.6%, which is lower than the GC-content (38%) of reported *E. faecium* genome (Lam et al., 2012). The genetic organization of the plasmid pSM409 is shown in Fig. 1. To confirm its circularity, the plasmid was digested with two different single cutter restriction endonucleases, *EcoRI* and *NcoI* (determined from restriction mapping) independently and was analysed through agarose gel electrophoresis. The linearized plasmid was predicted to be between 4 and 5 kb (Fig. 2). The plasmid pSM409 was found to contain six putative open reading frames and the level of similarity of each of the ORFs with the plasmids in the NCBI repository, identified by tBLASTn search, is shown in table 1. Sequence similarity search of plasmid pSM409 using megaBLAST against the NCBI nucleotide database and construction of Circos diagram with the top BLAST hits revealed a high degree of sequence heterogeneity and its mosaic nature as evidenced from the pattern of similarity of different local regions of the plasmid pSM409 to the plasmids in the database (Fig. 3) The maximum sequence similarity of 98.16% (54% of coverage) was obtained with the plasmid pCS17-A of *Streptococcus thermophilus* strain CS17. Like many other reported lactococcal theta plasmids (Émond et al., 2001), *repB* (ORF6), *orfX* (ORF5) and *hdsS* (ORF4) genes of pSM409 were found to exist in an operon, showing 83.01% similarity with the same of the plasmid p184E (CP034585.1), whereas the anti-toxin gene (ORF1) residing in a separate locus was found to be 99% similar to the same of pCRL291 (AF380336.1). ORF2 and ORF3 encoding the enterocin-A immunity protein and a hypothetical protein, respectively, were found to be 99.56% similar with that of pCS17-A (MF807212.1). The mosaic pattern of similarity predicts that the plasmid evolved as a consequence of recombination events (Górecki et al., 2011).

3.3. Replication and maintenance

In most plasmid replicons, plasmid-encoded initiator, RepB proteins are responsible for the initiation of plasmid replication. BLASTx comparison of ORF6 of plasmid pSM409 with the sequences available from NCBI database showed 90–91% identity with the RepB family plasmid

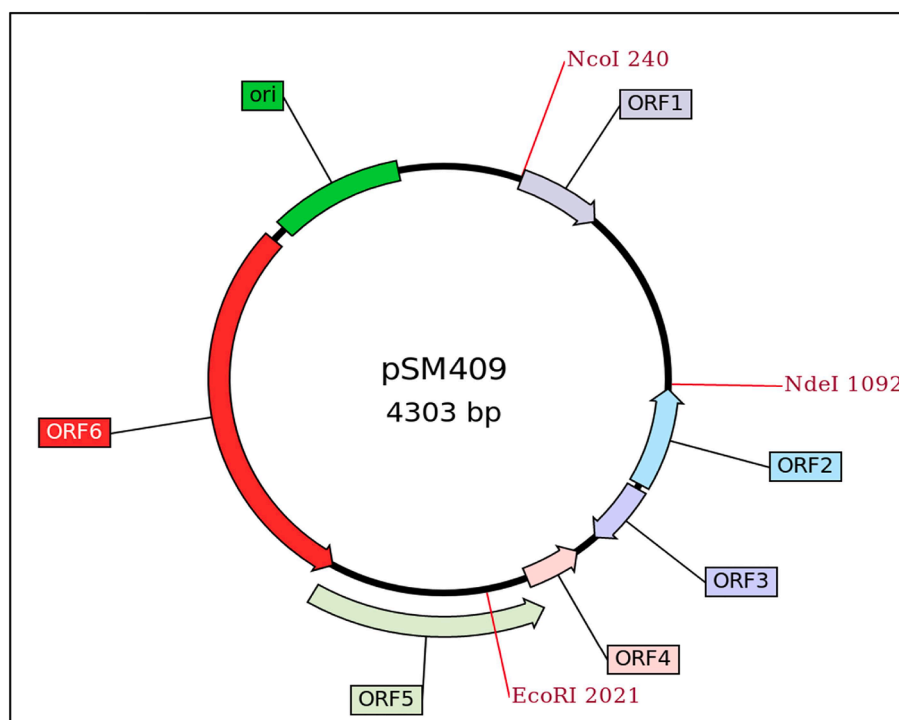


Fig. 1. Genetic map of the plasmid pSM409.

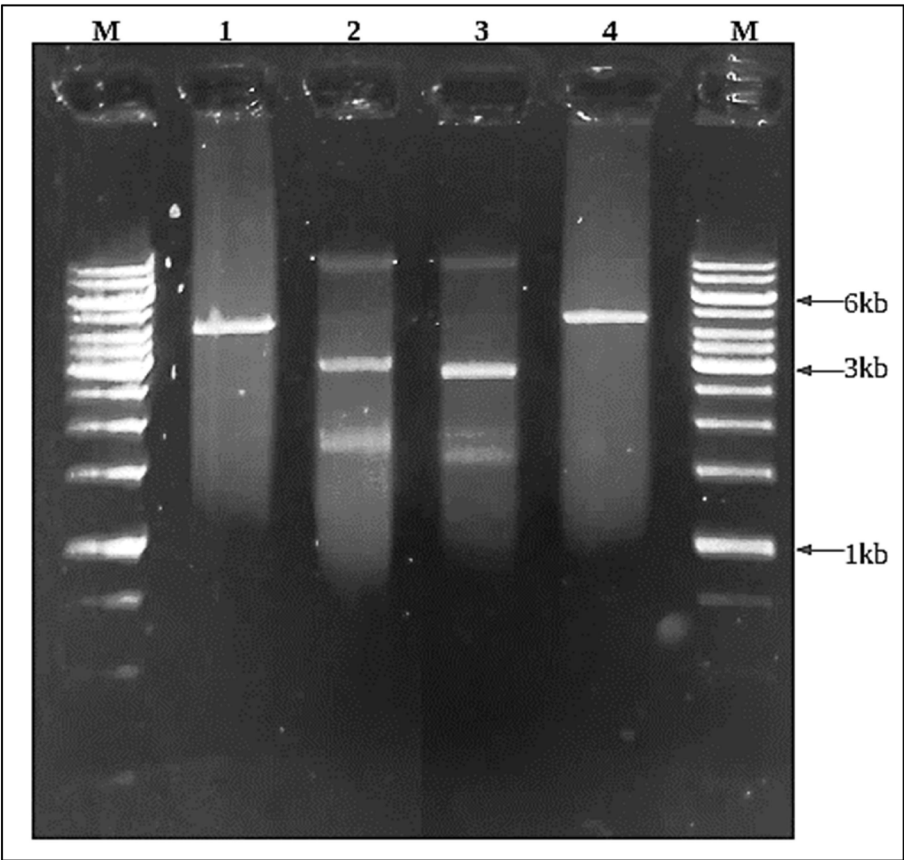


Fig. 2. Electropherogram of restriction digested pSM409 upon 1% agarose gel electrophoresis; lane M denotes the 1 kb DNA ladder (Fermentas), lane1 and 4 denotes *EcoRI* and *NcoI* digested plasmid pSM409, respectively, lane 2 and 3 denotes the undigested pSM409 plasmid.

Table 1
Bioinformatics analysis of the plasmid pSM409 sequence.

ORFs (open reading frame) in pSM409	Codon position	Putative Protein (DFAST Annotation)	Percent similarity as determined by tBLASTn search
ORF1	242–505	Antitoxin epsilon	80.46%, <i>Enterococcus faecalis</i> strain FC plasmid unnamed5, complete sequence CP028840.1
ORF2	1107–1427 *	Bacteriocin immunity protein	100.00%, <i>Streptococcus thermophilus</i> strain CS17 plasmid pCS17-A, complete sequence MF807212.1
ORF3	1443–1646 *	Hypothetical protein	97.01%, <i>Streptococcus thermophilus</i> strain CS17 plasmid pCS17-A, complete sequence MF807212.1
ORF4	1716–1892 *	HsdS:restriction endonuclease subunit S	91.43%, <i>Lactobacillus backii</i> strain TMW 1.1992 plasmid pL11992-2, complete sequence CP014892.1
ORF5	1880–2512 *	Hypothetical protein (ORFX)	88.57%, <i>Lactococcus lactis</i> subsp. <i>lactis</i> bv. <i>diacetylactis</i> plasmid pVF18, complete sequence JN172910.1
ORF6	2505–3716 *	Replication protein RepB	82.91%, <i>Enterococcus faecalis</i> strain ML21 plasmid PML21, complete sequence JX023543.1

* Complement.

replication initiator proteins of multiple species of *Enterococcus* (Accession WP_115236113.1, WP_002339159.1) and *Listeria monocytogenes* (WP_106509664.1, EAE8239723.1). Based on the comparative results, ORF6 encoding the RepB protein was predicted to be involved in the replication of the plasmid pSM409. BLAST search on the ACLAME database also suggested a role for the ORF6 in DNA replication initiation (go:0006270) and plasmid bidirectional (theta) vegetative DNA replication (phi:0000268). Further amino acid sequence analysis of RepB revealed the presence of two domains: Rep.3 superfamily (pfam01051) and the *L. lactis* RepB.C terminal superfamily (pfam06430) domains spanning between amino acid positions 26 to 263 and 266 to 387, respectively. A comparison of the nucleotide sequence of RepB of pSM409 with that of pCD4, a well-characterized theta-replicating lactococcal plasmid (Émond et al., 2001) revealed that the replication protein contained three signature motifs in the Rep3 superfamily domain: (i) a leucine zipper motif (amino acid positions 35 to 49) possibly involved in protein dimerization, (ii) a conserved domain (amino acid positions 133 to 148) believed to be involved in maintaining copy number and (iii) a 29 amino acid domain (positions 219 to 247) defined as the DNA binding site of several initiators and stretches of a 13-amino-acid (positions 220 to 232) proposed as the region governing *ori*-specific interactions.

To determine the level of similarity of ORF6 (RepB) of pSM409 with the previously reported circular plasmids, we employed a comparative study using tBLASTn program. The result revealed that the RepB shared 82.91% with the RepB proteins of *Enterococcus faecalis* strain PML21 plasmid PML21 (JX023543.1), 79.85% with *Corynebacterium ulcerans* strain 210931 plasmid pCul210931 (CP009584.1), and 76.43% identity with *Lactococcus lactis* subsp. *lactis* strain UC06 plasmid pUC06D (CP034579.1).

To gain further understanding of genetic diversity and phylogenetic

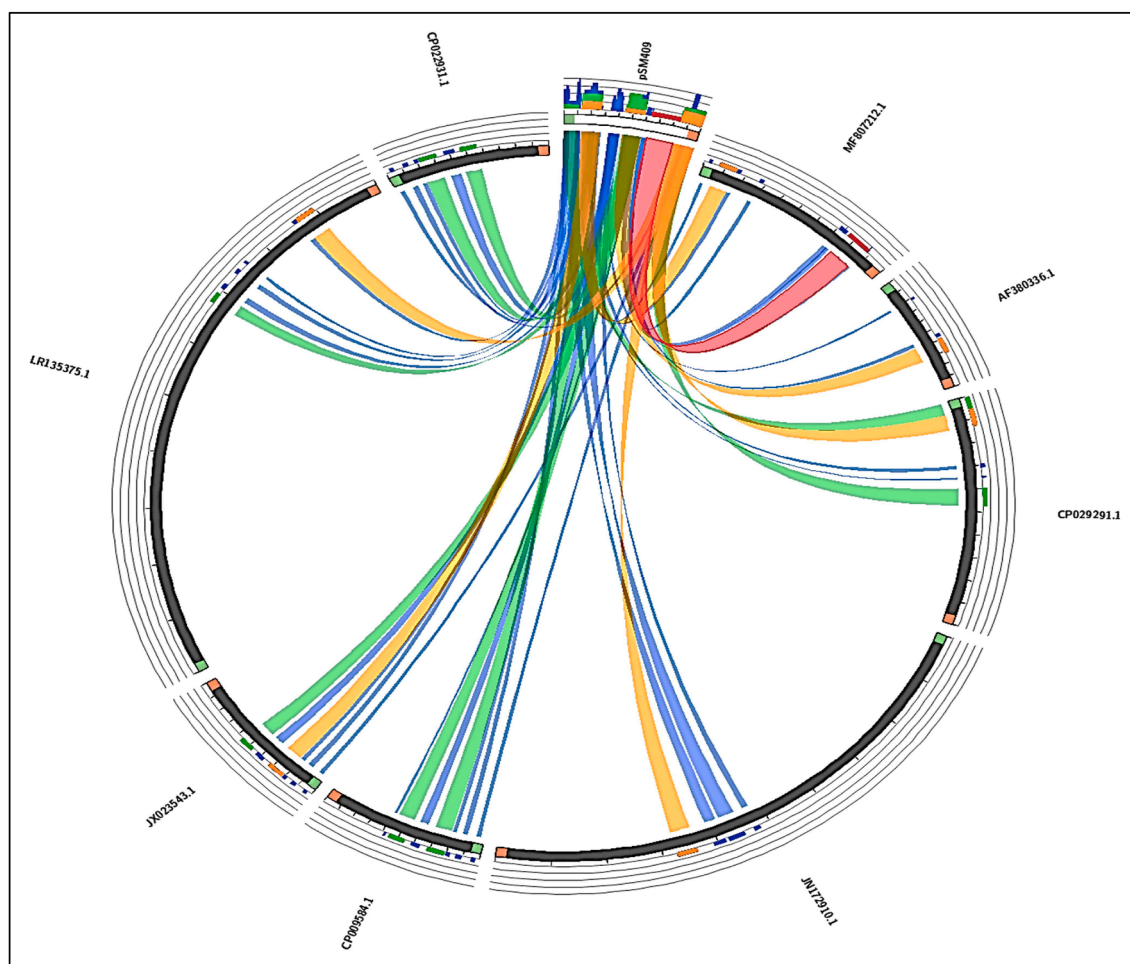


Fig. 3. Circos plot to visualize the sequence similarity of pSM409 using MF807212.1, CP022931.1, CP009584.1, JX023543.1, JN172910.1, LR135375.1, CP029291.1 and AF380336.1. The colours of ribbons/links represent BLAST similarity scores in the plot. The red colour signifies high sequence similarity followed by orange, green and blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

relationships of the replication initiator protein among theta plasmids, we retrieved the closest matching sequences from the NCBI database using tBLASTn and aligned with the amino acid sequence of the replication protein, RepB of pSM409. Based on this comparison, the DNA binding and the C-terminal domain of RepB of pSM409 were found to be variable when compared with RepB of PML21 (JX023543.1) and pCul210931 (CP009584.1), while the leucine zipper motif and the domain for copy number control remained to be identical (Fig. 4). The result suggests that essential domains required for replication are present in RepB of pSM409, but diverse in sequence. The phylogenetic analysis of pSM409-RepB protein showed that the replication initiation protein of pSM409 was clearly different and not affiliated to any of the clusters formed in the phylogenetic tree (Fig. 5), hence suggesting a novel type of replicon.

A comparison of the nucleotide sequence of pSM409 with the annotated plasmid sequences (Émond et al., 2001) indicated that the origin of replication of pSM409 is iteron-based and constituted of four and a half 22-bp imperfect direct repeats (iterons) followed by three inverted repeats of 5–7 bp (Fig. 6). Each of the direct repeats in the iterons of pSM409 was very similar to each other but differed by 1–2 nucleotides while the three 22-bp repeats of the iterons of reported plasmids (JX023543.1 and CP009584.1) are identical. When comparing the 22-bp iteron sequences of these two plasmids with pSM409, a conserved sequence patch (AAAAAAGTGTG) was found and the iterons of pSM409 differed from them by 5–7 nucleotides at the left half of each of the 22-bp repeats (Fig. 5). In upstream of the iteron there is an A-T rich region

with three direct repeats bracketed by the GC boxes, which is likely to be the recognition site for host-encoded functions involved in replication (Fig. 6). The direct repeating unit of the iteron is known to play a significant role in the initiation of plasmid replication by directly interacting with the replication protein. The iterons of similar RepB in the phylogenetic tree were found to be identical in different plasmids originated from different genera. The same is true for the DNA-binding domain. This suggests that there is a relationship between the iterons and the DNA-binding domain of the replication protein, for origin-specific DNA-protein interaction.

The ORF1 in pSM409 was annotated as antitoxin epsilon, which might contribute in enhanced stability to the plasmid in combination with the zeta toxin gene by post-segregational killing of plasmid-free cells as reported for other plasmids (Mruk and Kobayashi, 2014). BLASTx and CDS analysis of ORF1 at NCBI revealed that the encoded protein possesses a conserved Epsilon antitoxin domain (pfam08998), which shared 98.85%, 97.70% and 97.70% identity with antitoxin of multiple species of *Enterococcus* (WP_010824155.1, WP_010731447.1) and *Lactococcus lactis* (WP_010925612.1), respectively.

3.4. Accessory function

In addition to the essential genes encoding core plasmid function such as replication, the backbone of pSM409 also contained accessory genes like *hdsS* (ORF4), encoding one of the three subunits of type I restriction-modification system, and the ORF2 encoding a bacteriocin

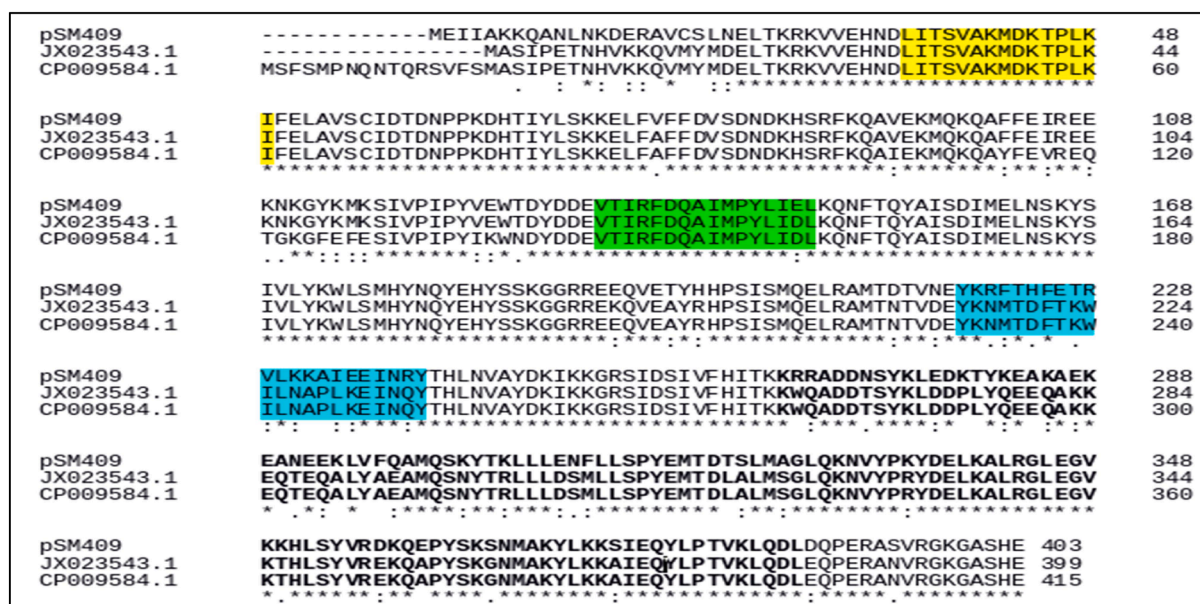


Fig. 4. ClustalW analysis of pSM409-RepB with RepB of *Enterococcus faecalis* strain PML21 plasmid PML21 (JX023543.1) and *Corynebacterium ulcerans* strain 210931 plasmid pCul210931 (CP009584.1). Yellow, green and blue highlighted regions mark the three conserved regions i.e. leucine zipper motif, copy number control motif and DNA-binding domain, respectively. The region marked in bold is the RepB_C terminal domain. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

immunity protein. The S subunit (HsdS subunit) of type 1 restriction-modification system recognizes specific DNA sequence for site-specific methylation and restriction activity (O'sullivan et al., 2000). The HsdS subunit determining the specificity of the methylase-endonuclease for DNA sequence recognition possibly mediates a protective role by restricting invading DNA (Górecki et al., 2011). A typical S-protein of type 1 restriction modification is of approximately 400 residues and consists of two conserved domains responsible for interaction with M-subunit and of two variable regions for sequence recognition (Górecki et al., 2011). However, HsdS of pSM409 was found to encode only 58 amino acids, hence, the HsdS without structural domains is truncated and is assumed to be non-functional. BLASTx analysis of ORF4 showed that the protein shared 89.47%, 91.43% and 72.34% identity with restriction endonuclease subunit S of *Lactococcus lactis* subsp. *lactis* (PFG89957.1), *Lactobacillus backii* (WP_083613290.1) and *Lactococcus lactis* (WP_003132926.1), respectively. ORF2 of pSM409 had 100%, 98.80% and 86.36% with the bacteriocin immunity proteins of *Lactobacillales* (WP_099390809.1), *Streptococcus thermophilus* (AUH26699.1) and *Enterococcus faecalis* (WP_080280360.1). The immunity protein of pSM409 contained an EntA_Immun (pfam08951) domain. Although the functional annotation was possible for four putative ORFs, no function could be predicted for ORF3 and ORF5 and thus the encoded product classified as hypothetical proteins.

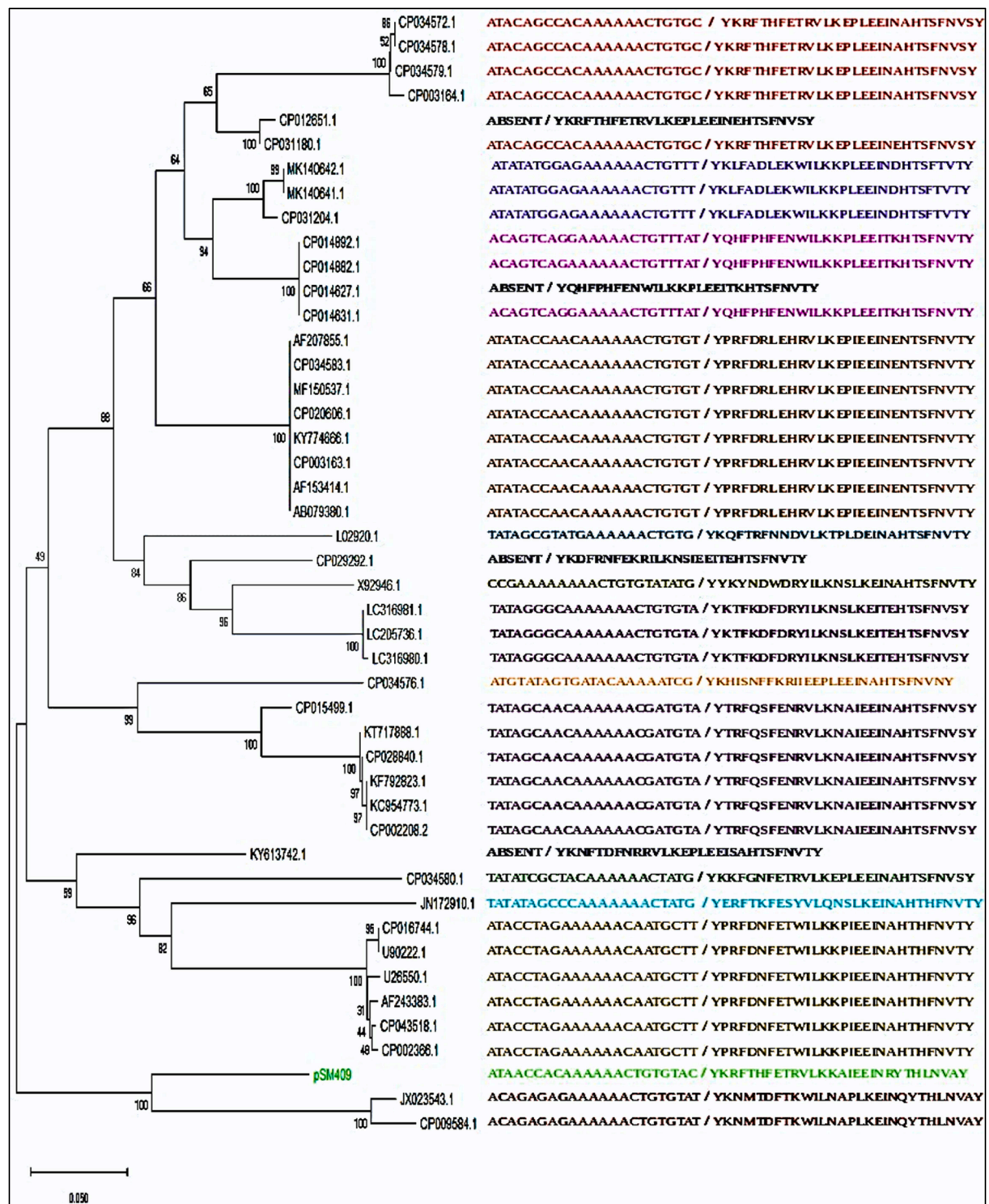
4. Discussion

The present study unravels a novel theta-replicating plasmid pSM409 of *Enterococcus faecium* RME isolated from raw milk. Plasmids are frequently found in enterococci and most of them carry antibiotic-resistance and virulent genes (Igbino and Beshiru, 2019; Mikalsen et al., 2015; Morroni et al., 2018). Although the plasmids containing the resistance and virulent genes create new interest in understanding the evolutionary mechanism of plasmid dissemination in terms of pathogenesis, they are not recommended by regulatory agencies for use in the development of genetic engineering tools due to potential safety concerns (Thomas, 2000). Hence, native plasmids devoid of such unwanted genes are important biological resources for applications in biotechnology. Sequence analysis of the plasmid pSM409 confirmed the

absence of such genes, but the plasmid was found to carry three accessory genes (37% of the whole plasmid) encoding a truncated HsdS subunit of type-1 restriction-modification system, antitoxin epsilon and bacteriocin immunity protein. Based on the previous study, these accessory genes are believed to be responsible for increased fitness in competitive environment (Górecki et al., 2011). Because of high degree of similarity of these genes with several plasmids from diverse genera, we speculate that the plasmid pSM409 is mosaic and evolved by acquisition of different fragments of DNA from different plasmids. The comparative analyses indicate that the plasmid pSM409 is completely new.

Like most other theta-replicating plasmids (Bruand et al., 1993; del Solar et al., 1998; Tabassum Khan, 2017), pSM409 contains a replication initiator protein RepB and an *ori* site encompassing the AT-rich region and a typical iteron sequence of 22-bp repeats. From homology search study using BLAST, we found high degree of similarity of RepB protein sequence of pSM409 with plasmid replication proteins from multiple species of different genera such as *Enterococcus*, *Listeria*, *Corynebacterium* and *Lactococcus*. Despite this similarity, the DNA binding domain and the RepB C-terminal domain were highly variable (Fig. 4). The pSM409-RepB appeared as distinctly different from other replication proteins clustering in the phylogenetic tree (Fig. 5). The result suggests that the plasmid pSM409 can be differentiated from the reported plasmids by domain-wise comparative analysis of RepB and also have a different variant of theta type origin.

In a previous study, with 25 Rep proteins of plasmids from lactic acid bacteria, it was reported that the 22-bp long repeats, the binding site of RepB, are diverse and unique to a similar type of replicons (Yao et al., 2019). In this study, we found a polarity in the distribution of dissimilar nucleotides in the 22-bp repeats of pSM409 with maximum variation observed within the first eight nucleotides in the repeats. It is known that monomeric Rep proteins interact throughout the length of the repeats for replication initiation through its DNA binding domain (Moscato et al., 1997). We established a relationship between the iterons and the DNA binding domain of plasmids shown in the tree (Fig. 5). The iterons in the same cluster as of Rep proteins are 100% identical, even though plasmids are originating from different genera. The same is true for DNA binding domain of RepB proteins. This can be explained by the



(caption on next page)

Fig. 5. Phylogenetic tree of RepB proteins using Neighbor-joining method with Mega X, The bootstrap values indicated at the nodes, were conferred from 1000 replicates. The optimal tree with the sum of branch length = 1.92254343 is shown. This analysis involved 46 RepB amino acid sequences from (JX023543.1) *Enterococcus faecalis* strain ML21 plasmid PML21; (CP009584.1) *Corynebacterium ulcerans* strain 210931 plasmid pCul210931; (CP034579.1) *Lactococcus lactis* subsp. *lactis* strain UC06 plasmid pUC06D; (CP034578.1) *Lactococcus lactis* subsp. *lactis* strain UC08 plasmid pUC08E; (CP034572.1) *Lactococcus lactis* subsp. *lactis* strain UC11 plasmid pCU11E; (CP031180.1) *Lactobacillus brevis* strain UCCLB556 plasmid pUCCLB556_F; (CP029292.1) *Lactococcus lactis* subsp. *lactis* KLD5 4.0325 plasmid unnamed5; (CP012651.1) *Lactobacillus plantarum* strain HFC8 plasmid pMK03; (CP034580.1) *Lactococcus lactis* subsp. *lactis* strain UC06 plasmid pUC06E; (CP003164.1) *Lactococcus lactis* subsp. *cremoris* UC509.9 plasmid pCIS2; (JN172910.1) *Lactococcus lactis* subsp. *lactis* *diacetylactis* plasmid pVF18; (MK140641.1) *Enterococcus faecalis* strain E035 plasmid pE035; (MK140642.1) *Enterococcus faecalis* strain E076 plasmid pE076; (CP031204.1) *Lactobacillus brevis* strain UCCLB5449 plasmid pUCCLB5449_F; (CP014892.1) *Lactobacillus backii* strain TMW 1.1992 plasmid pL11992-2; (CP014882.1) *Lactobacillus backii* strain TMW 1.1991 plasmid pL11991-1; (CP014631.1) *Lactobacillus backii* strain TMW 1.1988 plasmid L11988-7; (CP014627.1) *Lactobacillus backii* strain TMW 1.1988 plasmid L11988-3; (CP015499.1) *Lactobacillus helveticus* strain FAM22155 plasmid pFAM22155; (CP034576.1) *Lactococcus lactis* subsp. *cremoris* strain UC109 plasmid pUC109G; (L02920.1) *Lactococcus lactis* replication protein; (U26550.1) *Lactococcus lactis* replication protein; (CP043518.1) *Lactococcus lactis* subsp. *lactis* *diacetylactis* strain SD96 plasmid pSD96_06; (CP002366.1) *Lactococcus lactis* subsp. *lactis* CV56 plasmid pCV56A; (AF243383.1) *Lactococcus lactis* subsp. *lactis* plasmid pAH82; (CP016744.1) *Lactococcus lactis* subsp. *cremoris* strain JM2 plasmid pJM2C; (U90222.1) *Lactococcus lactis* plasmid pIL2614; (LC316981.1) *Lactococcus garvieae* plasmid pNs15soui29 DNA, strain: Ns15soui29; (LC205736.1) *Lactococcus garvieae* plasmid pBSLG13015 DNA, strain: BSLG13015; (KT717888.1) *Enterococcus faecalis* strain SZG10E plasmid pE30; (CP028840.1) *Enterococcus faecalis* strain FC plasmid unnamed5; (KF792823.1) *Enterococcus casseliflavus* strain En24 plasmid pEn24cfr; (KC954773.1) *Enterococcus faecalis* strain CPPF5 plasmid pCPPF5; (CP002208.2) *Enterococcus faecalis* strain EF-01 plasmid pEF-01; (LC316980.1) *Lactococcus garvieae* plasmid pNUF462; (X92946.1) *Lactococcus lactis* sp. *lactis* plasmid pK214; (CP034583.1) *Lactococcus lactis* subsp. *lactis* strain C10 plasmid pC10C; (MF150537.1) *Lactococcus lactis* subsp. *lactis* *diacetylactis* strain FM03P plasmid pLd9; (CP020606.1) *Lactococcus lactis* subsp. *lactis* *diacetylactis* strain FM03 plasmid pLd2; (KY774666.1) Cloning vector pCIE; (CP003163.1) *Lactococcus lactis* subsp. *cremoris* UC509.9 plasmid pCIS3; (AF153414.1) *Lactococcus lactis* plasmid pCIS3; (AB079380.1) *Lactococcus lactis* subsp. *lactis* *diacetylactis* cryptic plasmid pDR1-1B; (AF207855.1) *Lactococcus lactis* plasmid pAH33; (KY613742.1) *Listeria monocytogenes* strain LM-F-12 plasmid pLM-F-12. The iteron sequence and the associated DNA-binding domain of respective RepB protein is shown for each plasmids. Identical iterons and DNA-binding domains are marked with same colour code.



Fig. 6. The structural features of predicted origin of replication of pSM409. Blue arrows represent direct repeats, black discontinuous arrows are represented by inverted repeats, red-bold nucleotides are iteron region, black-bold nucleotides are A-T rich region and the orange-bold nucleotides are the G-C boxes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

fact that variation in the sequence of iterons diversifies the activity of RepB and as such each variant of RepB binds to a particular iteron sequence. However, experimental support is required to understand whether RepB from one cluster in the tree can initiate replication by interacting with iterons of another cluster. The phylogenetic data thus contribute to a new concept of classifying the theta-replicating plasmid based on the relationship of iterons and the DNA binding domain of RepB. The conventional system classified the theta plasmids on the basis of homology of replication proteins, while the present study pointed to the importance of both the replication protein and the iterons in classifying the theta plasmids, rendering in more simplicity and reliability of theta plasmid classification.

Based on the phylogenetic analysis, the structural similarity of RepB and the iterons of well-established theta plasmids, and the absence of other replication modules, the plasmid pSM409 is predicted to replicate via theta mechanism and the replicon is unique to pSM409. Over the rolling circle, theta type replicons are commonly preferred for ensuring

stability of plasmid-based expression system; hence, the pSM409-replicon offers an opportunity to be exploited for the construction of vectors for Gram-positive bacteria, for application in food and pharmaceutical industry.

The plasmid has evolved a mechanism for stable maintenance via post-segregational killing of plasmid-free cells through a putative toxin-antitoxin (TA) system. Plasmid-mediated TA system is widespread among bacteria and contributes stability to the plasmid (Meinhart et al., 2003; Romaniuk et al., 2017). The stable maintenance of the plasmid pSM409 was supported by a preliminary experiment in which *E. faecium* RME containing the plasmid pSM409 was sub-cultured in MRS-broth for a period of three years, no loss of plasmid was observed as inferred by the agarose gel electrophoresis (data not shown). However, TA-mediated stability of the pSM409 needs to be confirmed in mutant *E. faecium* RME lacking both the components of the TA system. Besides, the bacteriocin immunity protein was found to be present in the plasmid. This cytoplasmic immunity protein, unique for enterocin

(bacteriocin)-producing enterococci (Liu et al., 2019), is known to protect against an attack by bacteriocin-producing enterococci in the environment. No enterocin gene was found in the plasmid of the strain RME. The immunity protein encoded by the plasmid pSM409 is thus thought to provide cell survival benefit.

Taken together the results predict that the plasmid pSM409, which is different in architecture from the reported plasmids, may have evolved via recombination. Such evolutionary process that leads to formation of mosaic plasmids is widespread in nature (Peskesy et al., 2018).

5. Conclusion

The work presented herein reveals that the pSM409 isolated from *E. faecium* RME is a novel plasmid with mosaic architecture as evidenced by high degree of similarity of its different loci to different plasmids from diverse genera. The mosaic structure is associated with some important functions (toxin-antitoxin and restriction-modification system, and bacteriocin immunity) relevant to host cell survival. From a phylogenetic standpoint, the replication initiator protein RepB (91% identity with closest homologue) is distinct from other plasmids. The 22-bp repeats (iterons), the binding site of RepB, are structurally different as compared to reported plasmids. A comparative study and phylogenetic analysis reveal that the replication protein RepB of theta plasmids will have a distinct sequence variation in the DNA binding domain, if the 22-bp repeats are structurally different. It appears that the variable 22-bp repeat sequences possibly determine the variants of RepB initiator protein, they interact with. Hence, there exists a relationship between the 22-bp repeats and the DNA binding domain of RepB. It is thus a new concept of classifying theta plasmids based on this relationship. A unique sequence variation both in the repeats and the DNA binding domain of pSM409 suggests the pSM409 as a new theta plasmid. The plasmid pSM409 had remarkable stability under non-selective condition; it is small in size, theta type and devoid of genes for antibiotic-resistance or virulence factor. The pSM409 is thus, could be an attractive target for the development of advanced genetic engineering tools for enterococci. However, it necessitates further research

CRedit authorship contribution statement

Tawsif Ahmed Kazi: Conceptualization, Investigation, Methodology, Writing - original draft. **Suranjita Mitra:** Supervision, Writing - review & editing. **Bidhan Chandra Mukhopadhyay:** Data Curation. **Sukhendu Mandal:** Supervision, Writing - review & editing. **Swadesh Ranjan Biswas:** Project administration, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the Council of Scientific and Industrial Research (CSIR), Govt. of India, New Delhi (File No. 09/202(0070)/2017-EMR-I, Dated 25/10/2017) by providing Junior Research Fellowship (JRF)

References

Abouelnaga, M., Lamas, A., Quintela-Baluja, M., Osman, M., Miranda, J.M., Cepeda, A., Franco, C.M., 2016. Evaluation of the extent of spreading of virulence factors and antibiotic resistance in Enterococci isolated from fermented and unfermented foods. *Ann Microbiol* 66, 577–585. <https://doi.org/10.1007/s13213-015-1138-6>.

Altschul, S.F., Gish, W., Miller, W., Myers, E.W., Lipman, D.J., 1990. Basic local alignment search tool. *J. Mol. Biol.* 215, 403–410. [https://doi.org/10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2).

Arias, C.A., Contreras, G.A., Murray, B.E., 2010. Management of multidrug-resistant enterococcal infections. *Clin. Microbiol. Infect.* 16, 555–562. <https://doi.org/10.1111/j.1469-0691.2010.03214.x>.

Aziz, R.K., Bartels, D., Best, A.A., DeJongh, M., Disz, T., Edwards, R.A., Formsma, K., Gerdes, S., Glass, E.M., Kubal, M., Meyer, F., Olsen, G.J., Olson, R., Osterman, A.L., Overbeek, R.A., McNeil, L.K., Paarmann, D., Paczian, T., Parrello, B., Pusch, G.D., Reich, C., Stevens, R., Vassieva, O., Vonstein, V., Wilke, A., Zagnitko, O., 2008. The RAST server: rapid annotations using subsystems technology. *BMC Genomics* 9, 75. <https://doi.org/10.1186/1471-2164-9-75>.

Bankevich, A., Nurk, S., Antipov, D., Gurevich, A.A., Dvorkin, M., Kulikov, A.S., Lesin, V. M., Nikolenko, S.I., Pham, S., Pribelski, A.D., Pyshkin, A.V., Sirotkin, A.V., Vyahhi, N., Tesler, G., Alekseyev, M.A., Pevzner, P.A., 2012. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J. Comput. Biol.* 19, 455–477. <https://doi.org/10.1089/cmb.2012.0021>.

Ben Braïek, O., Smaoui, S., 2019. Enterococci: between emerging pathogens and potential probiotics. *Biomed Res. Int.* 5938210 <https://doi.org/10.1155/2019/5938210>.

Boetzer, M., Henkel, C.V., Jansen, H.J., Butler, D., Pirovano, W., 2011. Scaffolding pre-assembled contigs using SSPACE. *Bioinformatics* 27, 578–579. <https://doi.org/10.1093/bioinformatics/btq683>.

Bolger, A.M., Lohse, M., Usadel, B., 2014. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30, 2114–2120. <https://doi.org/10.1093/bioinformatics/btu170>.

Bruand, C., Le Chatelier, E., Ehrlich, S.D., Janniere, L., 1993. A fourth class of theta-replicating plasmids: The pAMβ1 family from Gram-positive bacteria. *Proc. Natl. Acad. Sci. U. S. A.* 90, 11668–11672. <https://doi.org/10.1073/pnas.90.24.11668>.

Camargo, C.H., Bruder-Nascimento, A., Lee, S.H.I., Fernandes Júnior, A., Kaneno, R., Rall, V.L.M., 2014. Prevalence and phenotypic characterization of Enterococcus spp. isolated from food in Brazil. *Braz. J. Microbiol.* 45, 111–115. <https://doi.org/10.1590/S1517-83822014000100016>.

Carasi, P., Jacquot, C., Romanin, D.E., Elie, A.-M., De Antoni, G.L., Urdaci, M.C., de los Serradell, M.A., 2014. Safety and potential beneficial properties of Enterococcus strains isolated from kefir. *Int. Dairy J.* 39, 193–200. <https://doi.org/10.1016/j.idairyj.2014.06.009>.

del Solar, G., Giraldo, R., Ruiz-Echevarría, M.J., Espinosa, M., Díaz-Orejías, R., 1998. Replication and control of circular bacterial plasmids. *Microbiol. Mol. Biol. Rev.* 62, 434–464. <https://doi.org/10.1128/MMBR.62.2.434-464.1998>.

Émond, É., Lavallée, R., Drolet, G., Moineau, S., LaPointe, G., 2001. Molecular characterization of a theta replication plasmid and its use for development of a two-component food-grade cloning system for Lactococcus lactis. *Appl. Environ. Microbiol.* 67, 1700–1709. <https://doi.org/10.1128/AEM.67.4.1700-1709.2001>.

Espinosa, M., del Solar, G., Rojo, F., Alonso, J.C., 1995. Plasmid rolling circle replication and its control. *FEMS Microbiol. Lett.* 130, 111–120. [https://doi.org/10.1016/0378-1097\(95\)00194-A](https://doi.org/10.1016/0378-1097(95)00194-A).

Felsenstein, J., 1985. Confidence limits on phylogenies: an approach using the bootstrap. *Evolution* 39, 783. <https://doi.org/10.2307/2408678>.

Franz, C.M.A.P., Huch, M., Abriouel, H., Holzapfel, W., Gálvez, A., 2011. Enterococci as probiotics and their implications in food safety. *Int. J. Food Microbiol.* 151, 125–140.

Gaglio, R., Couto, N., Marques, C., de Fatima Silva Lopes, M., Moschetti, G., Pomba, C., Settanni, L., 2016. Evaluation of antimicrobial resistance and virulence of enterococci from equipment surfaces, raw materials, and traditional cheeses. *Int. J. Food Microbiol.* 236, 107–114.

García-Solache, M., Rice, L.B., 2019. The Enterococcus: a model of adaptability to its environment. *Clin. Microbiol. Rev.* 32 <https://doi.org/10.1128/CMR.00058-18>.

Gilmore, M.S., Lebreton, F., van Schaik, W., 2013. Genomic transition of enterococci from gut commensals to leading causes of multidrug-resistant hospital infection in the antibiotic era. *Curr. Opin. Microbiol.* 16, 10–16.

Górecki, R.K., Koryszewska-Bagińska, A., Golebiewski, M., Zylńska, J., Grynberg, M., Bardowski, J.K., 2011. Adaptive potential of the lactococcus lactis IL594 strain encoded in its 7 plasmids. *PLoS One* 6. <https://doi.org/10.1371/journal.pone.0022238>.

Holzapfel, W., Arini, A., Aeschbacher, M., Coppolecchia, R., Pot, B., 2018. Enterococcus faecium SF68 as a model for efficacy and safety evaluation of pharmaceutical probiotics. *Benef. Microbes* 9, 375–388.

Igbinosa, E.O., Beshiru, A., 2019. Antimicrobial resistance, virulence determinants, and biofilm formation of Enterococcus species from ready-to-eat seafood. *Front. Microbiol.* 10 <https://doi.org/10.3389/fmicb.2019.00728>.

Ike, Y., 2017. Pathogenicity of Enterococci. *Nippon Saikingaku Zasshi* 72, 189–211. <https://doi.org/10.3412/jsb.72.189>.

Klibi, N., Aouini, R., Borgo, F., Ben Said, L., Ferrario, C., Dziri, R., Boudabous, A., Torres, C., Ben Slama, K., 2015. Antibiotic resistance and virulence of faecal enterococci isolated from food-producing animals in Tunisia. *Ann Microbiol* 65, 695–702.

Kumar, S., Stecher, G., Li, M., Knyaz, C., Tamura, K., 2018. MEGA X: molecular evolutionary genetics analysis across computing platforms. *Mol. Biol. Evol.* 35, 1547–1549. <https://doi.org/10.1093/molbev/msy096>.

Lam, M.M.C., Seemann, T., Bulach, D.M., Gladman, S.L., Chen, H., Haring, V., Moore, R. J., Ballard, S., Grayson, M.L., Johnson, P.D.R., Howden, B.P., Stinear, T.P., 2012. Comparative analysis of the first complete Enterococcus faecium genome. *J. Bacteriol.* 194, 2334–2341. <https://doi.org/10.1128/JB.00259-12>.

Langmead, B., Salzberg, S.L., 2012. Fast gapped-read alignment with Bowtie 2. *Nat. Methods* 9, 357–359. <https://doi.org/10.1038/nmeth.1923>.

- Lee, J.-H., Halgerson, J.S., Kim, J.-H., O'Sullivan, D.J., 2007. Comparative sequence analysis of plasmids from *Lactobacillus delbrueckii* and construction of a shuttle cloning vector. *Appl. Environ. Microbiol.* 73, 4417–4424. <https://doi.org/10.1128/AEM.00099-07>.
- Lee, T., Pang, S., Abraham, S., Coombs, G.W., 2019. Antimicrobial-resistant CC17 *Enterococcus faecium*: The past, the present and the future. *J. Glob. Antimicrob. Resist.* 16, 36–47. <https://doi.org/10.1016/j.jgar.2018.08.016>.
- Liu, G., Wang, Y., Li, X., Hao, X., Xu, D., Zhou, Y., Mehmood, A., Wang, C., 2019. Genetic and biochemical evidence that *Enterococcus faecalis* Gr17 produces a novel and sec-dependent bacteriocin, Enterocin Gr17. *Front. Microbiol.* 10 <https://doi.org/10.3389/fmicb.2019.01806>.
- Machón, C., Espinosa, M., Ruiz-Masó, J.A., Del Solar, G., Bordanaba-Ruiseco, L., Coll, M., 2015. Plasmid rolling-circle replication. *Microbiol. Spectr.* <https://doi.org/10.1128/microbiolspec.PLAS-0035-2014>.
- Meinhart, A., Alonso, J.C., Strater, N., Saenger, W., 2003. Crystal structure of the plasmid maintenance system epsilon/zeta: functional mechanism of toxin zeta and inactivation by epsilon 2 zeta 2 complex formation. *Proc. Natl. Acad. Sci. U. S. A.* 100, 1661–1666. <https://doi.org/10.1073/pnas.0434325100> [pii].
- Mierau, I., Kleerebezem, M., 2005. 10 years of the nisin-controlled gene expression system (NICE) in *Lactococcus lactis*. *Appl. Microbiol. Biotechnol.* 68, 705–717.
- Mikalsen, T., Pedersen, T., Willems, R., Coque, T.M., Werner, G., Sadowy, E., van Schaik, W., Jensen, L.B., Sundsfjord, A., Hegstad, K., 2015. Investigating the mobilome in clinically important lineages of *Enterococcus faecium* and *Enterococcus faecalis*. *BMC Genomics* 16. <https://doi.org/10.1186/s12864-015-1407-6>.
- Mitra, S., Chakrabarty, P.K., Biswas, S.R., 2005. Production and characterization of Nisin-like peptide produced by a strain of *Lactococcus lactis* isolated from fermented milk. *Curr. Microbiol.* 51, 183–187.
- Morroni, G., Brenciani, A., Antonelli, A., Maria D'Andrea, M., Di Pilato, V., Fioriti, S., Mingoia, M., Vignaroli, C., Cironi, O., Biavasco, F., Varaldo, P.E., Rossolini, G.M., Giovanetti, E., 2018. Characterization of a multiresistance plasmid carrying the *optrA* and *cfr* resistance genes from an *Enterococcus faecium* clinical isolate. *Front. Microbiol.* 9 <https://doi.org/10.3389/fmicb.2018.02189>.
- Moscato, M., Eritja, R., Espinosa, M., 1997. Initiation of replication of plasmid pMV158: mechanisms of DNA strand-transfer reactions mediated by the initiator RepB protein. *J. Mol. Biol.* 268, 840–856.
- Mruk, I., Kobayashi, I., 2014. To be or not to be: regulation of restriction–modification systems and other toxin–antitoxin systems. *Nucleic Acids Res.* 42, 70–86. <https://doi.org/10.1093/nar/gkt711>.
- Murray, B.E., 1990. The life and times of the *Enterococcus*. *Clin. Microbiol. Rev.* 3, 46–65. <https://doi.org/10.1128/CMR.3.1.46-65.1990>.
- O'Sullivan, D., Twomey, D.P., Coffey, A., Hill, C., Fitzgerald, G.F., Ross, R.P., 2000. Novel type I restriction specificities through domain shuffling of HsdS subunits in *Lactococcus lactis*. *Mol. Microbiol.* 36, 866–875.
- Özmen Toğay, S., Çelebi Keskin, A., Açı, L., Temiz, A., 2010. Virulence genes, antibiotic resistance and plasmid profiles of *Enterococcus faecalis* and *Enterococcus faecium* from naturally fermented Turkish foods. *J. Appl. Microbiol.* 109, 1084–1092. <https://doi.org/10.1111/j.1365-2672.2010.04763.x>.
- Peskesy, M., Tilley, R., Beck, D., 2018. Mosaic plasmids are abundant and unevenly distributed across prokaryotic taxa. Mosaic plasmids are Abund. unevenly Distrib. across prokaryotic taxa. <https://doi.org/10.1101/428219>.
- Romaniuk, K., Krucon, T., Decewicz, P., Gorecki, A., Dziewit, L., 2017. Molecular characterization of the pA3J1 plasmid from the psychrotolerant Antarctic bacterium *Pseudomonas* sp. ANT_J3. *Plasmid* 92, 49–56. <https://doi.org/10.1016/j.plasmid.2017.08.001>.
- Saitou, N., Nei, M., 1987. The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Mol. Biol. Evol.* 4, 406–425. <https://doi.org/10.1093/oxfordjournals.molbev.a040454>.
- Tabassum Khan, N., 2017. Mechanisms of plasmid replication. *J. Proteomics Bioinform.* 10 <https://doi.org/10.4172/jpb.1000444>.
- Thomas, C.M., 2000. Paradigms of plasmid organization. *Mol. Microbiol.* 37, 485–491. <https://doi.org/10.1046/j.1365-2958.2000.02006.x>.
- Tomita, H., Pierson, C., Lim, S.K., Clewell, D.B., Ike, Y., 2002. Possible connection between a widely disseminated conjugative gentamicin resistance (pMG1-Like) plasmid and the emergence of vancomycin resistance in *Enterococcus faecium*. *J. Clin. Microbiol.* 40, 3326–3333.
- Tomita, H., Tanimoto, K., Hayakawa, S., Morinaga, K., Ezaki, K., Oshima, H., Ike, Y., 2003. Highly conjugative pMG1-Like plasmids carrying Tn1546-Like transposons that encode vancomycin resistance in *Enterococcus faecium*. *JB* 185, 7024–7028. <https://doi.org/10.1128/JB.185.23.7024-7028.2003>.
- Wu, Y., Zhen, W., Geng, Y., Wang, Z., Guo, Y., 2019. Pretreatment with probiotic *Enterococcus faecium* NCIMB 11181 ameliorates necrotic enteritis-induced intestinal barrier injury in broiler chickens. *Sci. Rep.* 9 <https://doi.org/10.1038/s41598-019-46578-x>.
- Yao, F., Xu, X., Du, X., Cao, K., Pan, Q., 2019. Detection and characterization of a theta-replicating plasmid pLP60 from *Lactobacillus plantarum* PC518 by inverse PCR. *Heliyon* 5, e02164. <https://doi.org/10.1016/j.heliyon.2019.e02164>.
- Yuksel, F.N., Akcelik, N., Akcelik, M., 2015. Incidence of antibiotic resistance and virulence determinants in *Enterococcus faecium* and *Enterococcus faecalis* strains, isolated from traditional cheeses in Turkey. *Mol. Genet. Microbiol. Virol.* 30, 206–215. <https://doi.org/10.3103/S089141681504014X>.
- Zerbino, D.R., Birney, E., 2008. Velvet: Algorithms for de novo short read assembly using de Bruijn graphs. *Genome Res.* 18, 821–829. <https://doi.org/10.1101/gr.074492.107>.
- Zuckerkandl, E., Pauling, L., 1965. Evolutionary divergence and convergence in proteins. In: *Evolving Genes and Proteins*. Elsevier, pp. 97–166. <https://doi.org/10.1016/B978-1-4832-2734-4.50017-6>.